Timothy Hansen

612-453-7312 | t@tjhansen.net | linkedin.com/in/timothy-hansen-dev | github.com/TimothyH3

Education

Arizona State University Tempe, AZ
Bachelor of Science in Software Engineering *Expected Dec 2026*

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Software Enterprise: Construction & Transition, Database Management Systems, Web & Cloud Applications, Logic in Computer Science, Object-Oriented Design

Technical Skills

Languages: Java, TypeScript, JavaScript, Python, SQL (PostgreSQL, MySQL), C, C#
Frontend & Web: Vue 3 (Composition API), Pinia, Vite, HTML5, CSS3, Leaflet, OpenStreetMap, REST APIs
Backend & Databases: Node.js, Express.js, PostgreSQL, MySQL, Relational Database Normalization, DDL/DML
Testing & Quality Assurance: JUnit 5, Vitest, JaCoCo, SpotBugs, Checkstyle, Static Analysis, Unit Testing
Tools & Methodologies: Git, GitHub Actions, Agile/Scrum (Scrum Master), CI/CD, Linux/Unix CLI, GoF Design Patterns

Technical Projects

Come Together Compass | *Vue 3, TypeScript, Node.js, Express, PostgreSQL, Pinia, Vitest* Aug 2026 – Present

- Architecting a single-page community resource platform using Vue 3 (Composition API), TypeScript, and Pinia, backed by a Node.js/Express REST API and PostgreSQL to connect individuals with verified local aid organizations.
- Integrated Leaflet and OpenStreetMap with the U.S. Census Bureau Geocoding API to dynamically map local community resources, eliminating recurring commercial map licensing costs and third-party API quotas.
- Engineered a privacy-first emotional support tool executing strictly in browser RAM via reactive Pinia stores to guarantee complete session anonymity, featuring automated triage escalations to the 988 Crisis Lifeline.
- Built automated unit test suites using Vitest to validate reactive state transitions and data repositories, integrated into a continuous integration pipeline via GitHub Actions on pull requests.

Healthcare Operations Simulation | *Java, JUnit 5, JaCoCo, SpotBugs, Checkstyle* Oct 2025 – Dec 2025

- Engineered an event-driven Java simulation using Facade, Command, and Factory Method patterns, decoupling clinical action triggers from state execution to eliminate monolithic control logic.
- Centralized randomized patient demographic generation and insurance tier billing rules into dedicated factory modules, streamlining isolated unit test execution.
- Achieved 87% branch and statement test coverage via JUnit 5 and JaCoCo, leveraging Java Reflection to assert private state transitions and intercepting standard I/O streams to validate audit logging.
- Enforced enterprise static code analysis standards, achieving zero-defect compliance across Checkstyle and SpotBugs rulesets.

Martial Arts Gym Scheduling Platform | *Java, Swing/AWT, Git, GitHub, Agile/Scrum* Oct 2025 – Dec 2025

- Served as Scrum Master for an Agile team, managing sprint planning, backlog grooming across 11 user stories, and daily standups to sustain velocity through sudden team downsizing.
- Refactored *Memoranda* (an open-source Java desktop platform), isolating legacy UI dependencies to build a modular, customized gym scheduling interface.
- Engineered an icon helper and asset rendering utility to resolve legacy display scaling issues and aspect-ratio distortion across varying OS display resolutions.
- Standardized GitHub pull request workflows and branch merge criteria, conducting peer code reviews to prevent regressions and maintain build stability.